

SQL & DATABASES:

NEW WHEELS PROJECT REPORT

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BUSINESS CONTEXT

A lot of people in the world share a common desire: to own a vehicle. A car or an automobile is seen as an object that gives the freedom of mobility. Many are now preferring pre-owned vehicles because they come at an affordable cost, but at the same time, they are also concerned about whether the after-sales service provided by the resale vendors is as good as the care you may get from the actual manufacturers. New-Wheels, a vehicle resale company, has launched an app with an end-to-end service from listing the vehicle on the platform to shipping it to the customer's location. This app also captures the overall after-sales feedback given by the customer.

PROBLEM STATEMENT

New-Wheels sales have been dipping steadily in the past year, and due to the critical customer feedback and ratings online, there has been a drop in new customers every quarter, which is concerning to the business. The CEO of the company now wants a quarterly report with all the key metrics sent to him so he can assess the health of the business and make the necessary decisions.



TOTAL REVENUE

82.6 M

TOTAL ORDERS

1000

TOTAL CUSTOMERS

994

AVERAGE RATING

3.07

LAST QTR REVENUE

15.1 M

LAST QTR ORDERS

199

AVERAGE DAYS TO SHIP

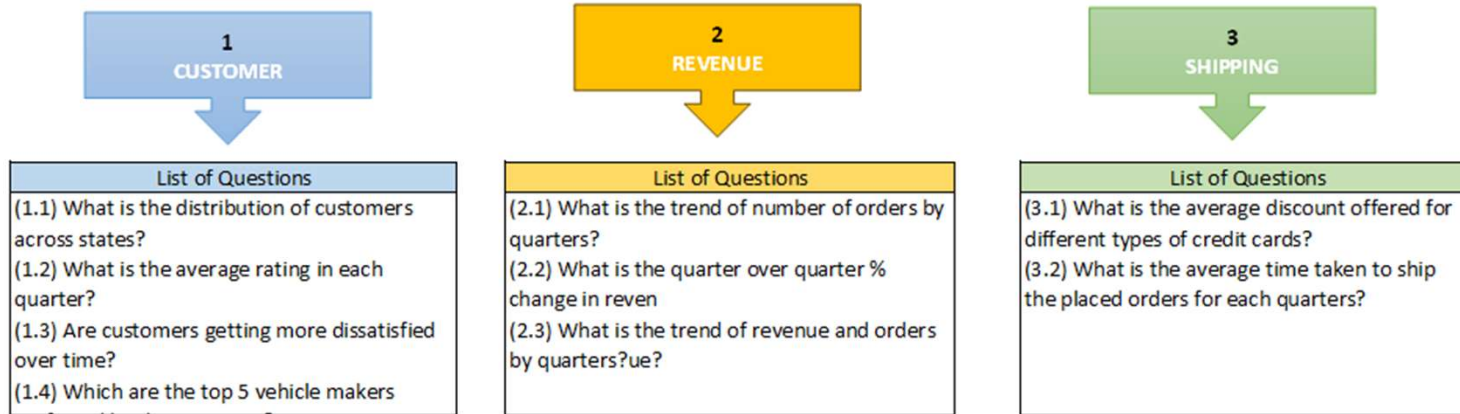
105

AVERAGE % OF GOOD
FEEDBACK

20.47%

BUSINESS OVERVIEW

What are the facts that affecting the new wheels business?

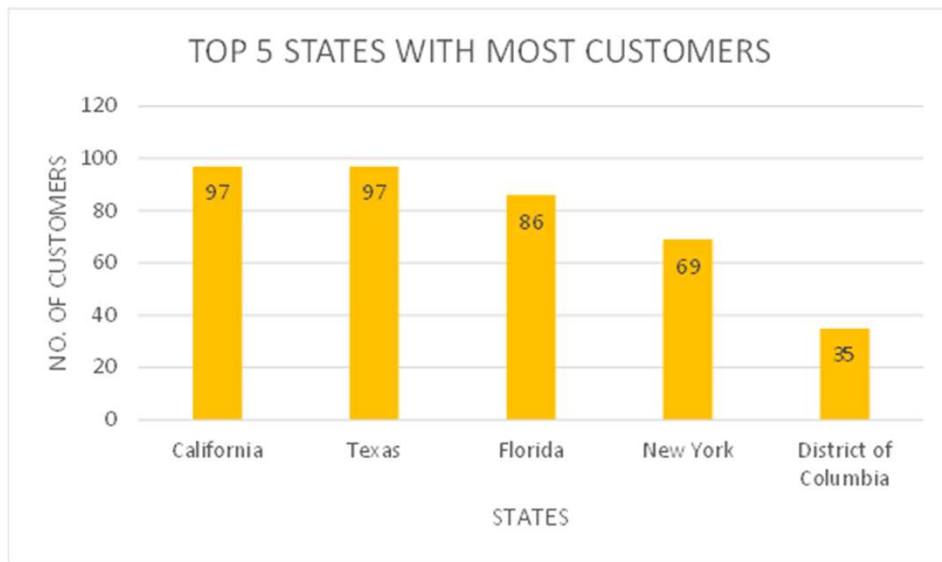


METRICS INSIGHT

CUSTOMER METRICS



DISTRIBUION OF CUSTOMERS ACROSS STATES

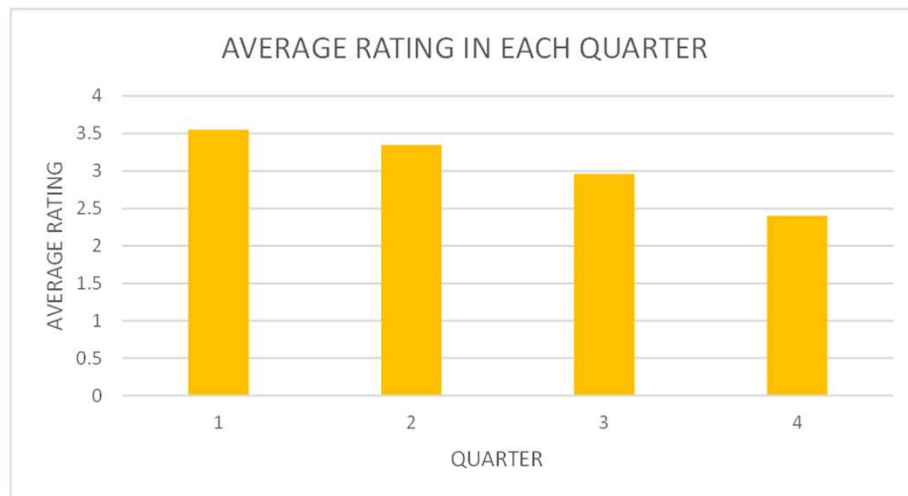


state	number_of_customers
California	97
Texas	97
Florida	86
New York	69
District of Columbia	35

OBVERSATIONS/FINDINGS

- There are a total of 49 states with a customer base of 994.
- The top-5 states with the most customers are California, Texas, Florida, New York, and the District of Columbia, accounting for 38.6% of the total customers.
- On the other hand, the 5 states with the least customers are Vermont (1), Wyoming (1), Maine (1), Mississippi (2), and North Dakota (2), making up less than 1% of the total customer base.

AVERAGE CUSTOMERS RATINGS BY QUARTER

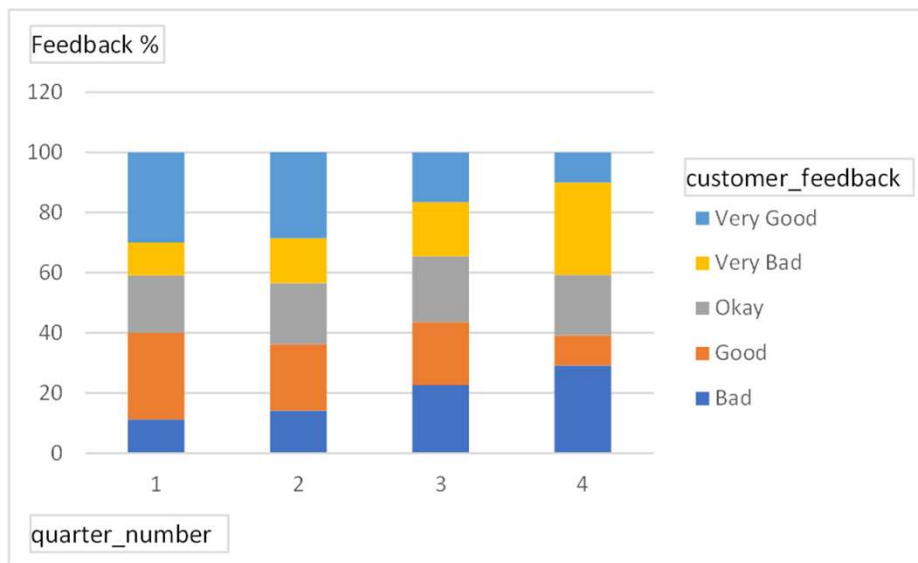


quarter_number	avg_rating
1	3.55
2	3.35
3	2.96
4	2.40

OBVERSATIONS/FINDINGS

- According to the data, the average rating for quarter 1 is 3.55, quarter 2 is 3.3, quarter 3 is 2.96, and quarter 4 is 2.40.
- The average rating is declining from quarter to quarter.
- The rating trend indicates a negative shift in customer satisfaction when comparing from quarter 1 to quarter 4.

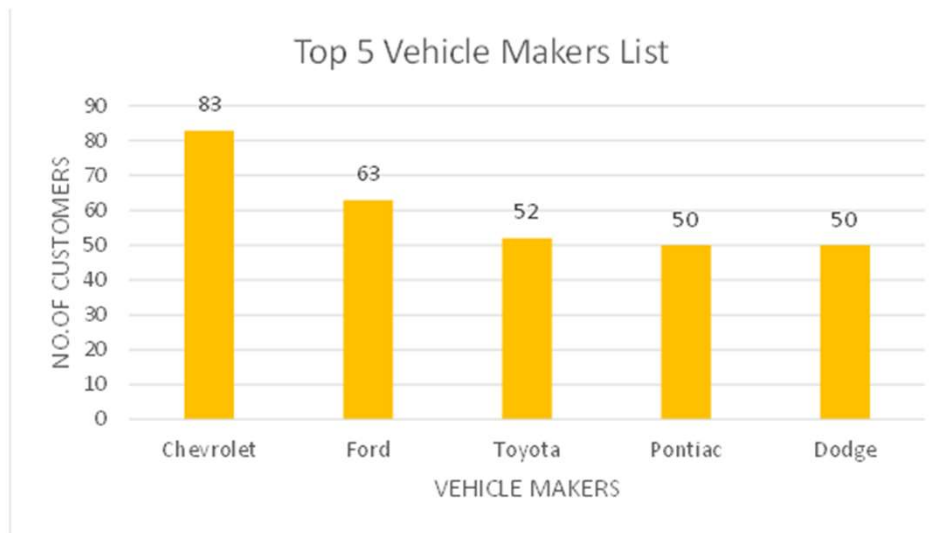
TREND OF CUSTOMER SATISFACTION



OBVERSATIONS/FINDINGS

- Out of a total of 1000 customer feedback responses, the breakdown is as follows: very good - 226, good - 215, neutral - 202, bad – 182, and very bad - 175.
- In quarter 1, the percentage of very good feedback was 30%, but it declined to 10% in quarter 4.
- The combined percentage of very good and good was 58.71% in quarter 1, but it declined to 20.1% in quarter 4.
- Similarly, in quarter 1, the percentage of very bad feedback was 10.97%, but it increased to 30.65% in quarter 4.
- The combined percentage of very bad and bad was 22.26% in quarter 1, but it increased to 59.8% in quarter 4.
- The data suggests a growing dissatisfaction among customers over time.

TOP VEHICLE MAKERS PREFERRED BY CUSTOMERS



vehicle_maker	total_customer
Chevrolet	83
Ford	63
Toyota	52
Pontiac	50
Dodge	50

OBVERSATIONS/FINDINGS

- Out of a total of 54 vehicle makers, Chevrolet is the top choice among customers.
- Following Chevrolet, Ford, Toyota, Pontiac, and Dodge make up the top-5 list.
- These top-5 vehicle makers collectively hold around 30% of the market share of total customers.

MOST PREFERRED VEHICLE MAKE IN EACH STATE

OBVERSATIONS/FINDINGS

- Chevrolet is the most preferred vehicle maker in many states, with a total of 17 out of 49 states preferring Chevrolet.
- Following Chevrolet, Dodge is preferred in 12 states, and Ford in 10 states.
- The least preferred vehicle makers include Chrysler, Ferrari, Oldsmobile, Ram and Saab.

state	vehicle_maker	no_of_customers
Alabama	Dodge	5
Alaska	Chevrolet	2
Arizona	Cadillac	3
Arizona	Pontiac	3
Arkansas	Chevrolet	1
Arkansas	GMC	1
Arkansas	Mitsubishi	1
Arkansas	Pontiac	1
Arkansas	Suzuki	1
Arkansas	Volkswagen	1
California	Audi	6
California	Chevrolet	6
California	Dodge	6
California	Ford	6
California	Nissan	6
Colorado	Chevrolet	5
Connecticut	Chevrolet	2
Connecticut	Maserati	2
Connecticut	Mercury	2
Connecticut	Volvo	2
Delaware	Mitsubishi	2
District of Columbia	Chevrolet	4
Florida	Toyota	7
Georgia	Toyota	3
Hawaii	Cadillac	1
Hawaii	Ford	1
Hawaii	GMC	1
Hawaii	Nissan	1
Hawaii	Pontiac	1
Hawaii	Toyota	1
Idaho	Dodge	2
Illinois	Chevrolet	3
Illinois	GMC	3
Illinois	Ford	3
Indiana	Mazda	4

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state	vehicle_maker	no_of_customers
Iowa	Chevrolet	1
Iowa	Chrysler	1
Iowa	Dodge	1
Iowa	Ford	1
Iowa	Hyundai	1
Iowa	Isuzu	1
Iowa	Jeep	1
Iowa	Mazda	1
Iowa	Pontiac	1
Iowa	Porsche	1
Iowa	Subaru	1
Kansas	Buick	1
Kansas	Dodge	1
Kansas	Ford	1
Kansas	GMC	1
Kansas	Honda	1
Kansas	Lexus	1
Kansas	Maserati	1
Kansas	Mazda	1
Kansas	Mercedes-Benz	1
Kansas	Nissan	1
Kansas	Saab	1
Kansas	Suzuki	1
Kansas	Volkswagen	1
Kentucky	Acura	1
Kentucky	Audi	1
Kentucky	Mercedes-Benz	1
Kentucky	Mercury	1
Kentucky	Nissan	1
Kentucky	Pontiac	1
Kentucky	Ram	1
Kentucky	Volvo	1
Louisiana	Nissan	2
Louisiana	BMW	2
Louisiana	Ford	2
Louisiana	Pontiac	2
Louisiana	Kia	2
Maine	Mercedes-Benz	1
Maryland	Ford	5
Massachusetts	Dodge	2
Massachusetts	Chevrolet	2
Michigan	Ford	3
Minnesota	GMC	3

state	vehicle_maker	no_of_customers
Mississippi	Dodge	1
Mississippi	Toyota	1
Missouri	Chevrolet	4
Montana	Chevrolet	1
Montana	Dodge	1
Montana	Mitsubishi	1
Nebraska	Cadillac	1
Nebraska	Chevrolet	1
Nebraska	Mercedes-Benz	1
Nebraska	Nissan	1
Nebraska	Pontiac	1
Nebraska	Toyota	1
Nebraska	Volkswagen	1
Nevada	Pontiac	3
New Hampshire	Chrysler	1
New Hampshire	Lexus	1
New Hampshire	Lincoln	1
New Jersey	Hyundai	2
New Jersey	Mercedes-Benz	2
New Mexico	Dodge	2
New York	Toyota	5
New York	Pontiac	5
North Carolina	Volvo	3
North Dakota	Ford	1
North Dakota	Hyundai	1
Ohio	Chevrolet	6
Oklahoma	Ferrari	2
Oklahoma	Mazda	2
Oklahoma	Toyota	2
Oregon	Toyota	2
Pennsylvania	Toyota	3
South Carolina	Acura	1
South Carolina	BMW	1
South Carolina	Buick	1
South Carolina	Dodge	1
South Carolina	Isuzu	1
South Carolina	Jaguar	1
South Carolina	Kia	1
South Carolina	Mazda	1
South Carolina	Mitsubishi	1
Tennessee	Mazda	3
Texas	Chevrolet	9

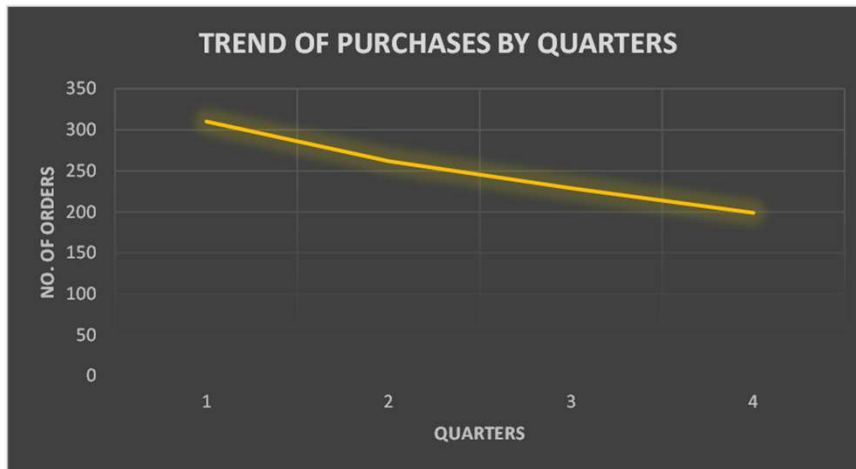
state	vehicle_maker	no_of_customers
Utah	Buick	1
Utah	Chevrolet	1
Utah	Dodge	1
Utah	Isuzu	1
Utah	Lincoln	1
Utah	Maybach	1
Utah	Oldsmobile	1
Utah	Pontiac	1
Utah	Subaru	1
Utah	Volkswagen	1
Vermont	Mazda	1
Virginia	Ford	5
Washington	Chevrolet	5
West Virginia	Mercedes-Benz	2
Wisconsin	Acura	1
Wisconsin	Cadillac	1
Wisconsin	Chevrolet	1
Wisconsin	Dodge	1
Wisconsin	Honda	1
Wisconsin	Mazda	1
Wisconsin	Nissan	1
Wisconsin	Pontiac	1
Wyoming	Buick	1

*MOST PREFERRED
VEHICLE MAKE IN
EACH STATE*

REVENUE METRICS



TREND OF PURCHASE BY QUARTER

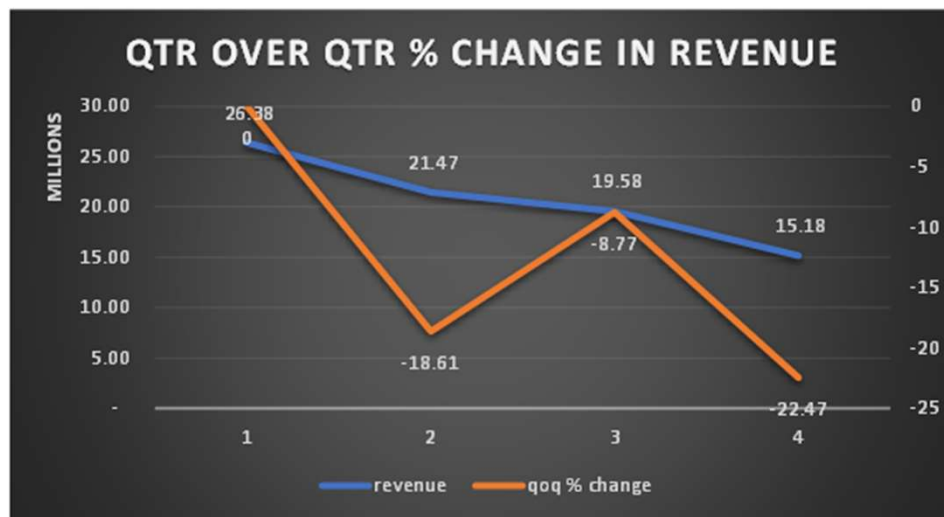


quarter_number	no_of_orders
1	310
2	262
3	229
4	199

OBVERSATIONS/FINDINGS

- Orders are decreasing for all quarters.
- In percentage terms, the customer purchasing rate decreasing from 31% in quarter 1 to 20% in quarter 4.

QUARTER ON QUARTER % CHANGE IN REVENUE

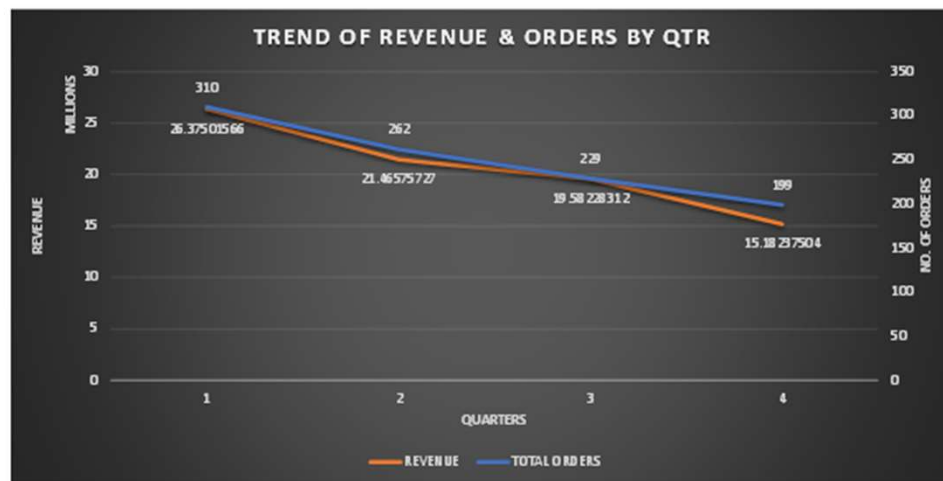


quarter_number	revenue	previous_revenue	qoq % change
1	26,375,015.66	NULL	NULL
2	21,465,757.27	26,375,015.66	-18.61
3	19,582,283.12	21,465,757.27	-8.77
4	15,182,375.04	19,582,283.12	-22.47

OBVERSATIONS/FINDINGS

- The revenue for each quarter is as follows: quarter 1 - 26 million, quarter 2 - 21 million, quarter 3 - 19 million, and quarter 4 - 15 mil.
- The revenue change in quarter 2 is -18%, in quarter 3 is -8%, and in quarter 4 is -22%.
- The revenue change ranges from -18% to -22% across all quarters.
- According to the data, revenue is decreasing in each quarter.

TREND OF REVENUE AND ORDERS BY QUARTER



OBSERVATIONS/FINDINGS

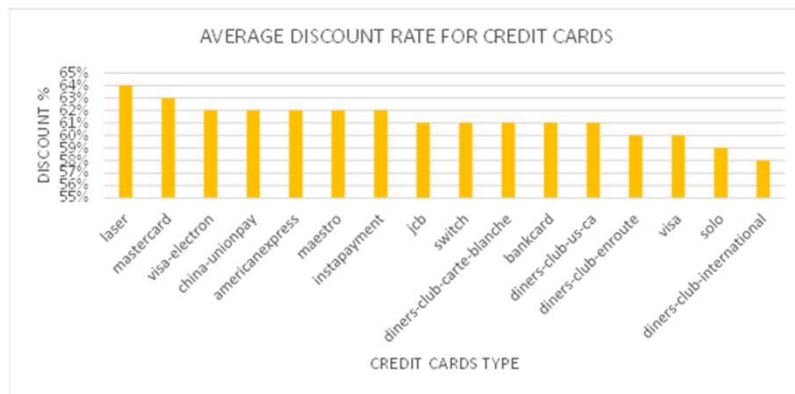
- In quarter 1, the revenue was 26 million, with a total order of 310 orders, indicating that the business performed well in that quarter.
- Starting from quarter 2, both revenue and the number of orders have been decreasing each quarter.
- There is an 11% change in total orders from quarter 1 to quarter 4.

quarter_number	revenue	total_orders
1	26375015.66	310
2	21465757.27	262
3	19582283.12	229
4	15182375.04	199



SHIPPING METRICS

AVERAGE DISCOUNT OFFERED BY CREDIT CARD TYPE

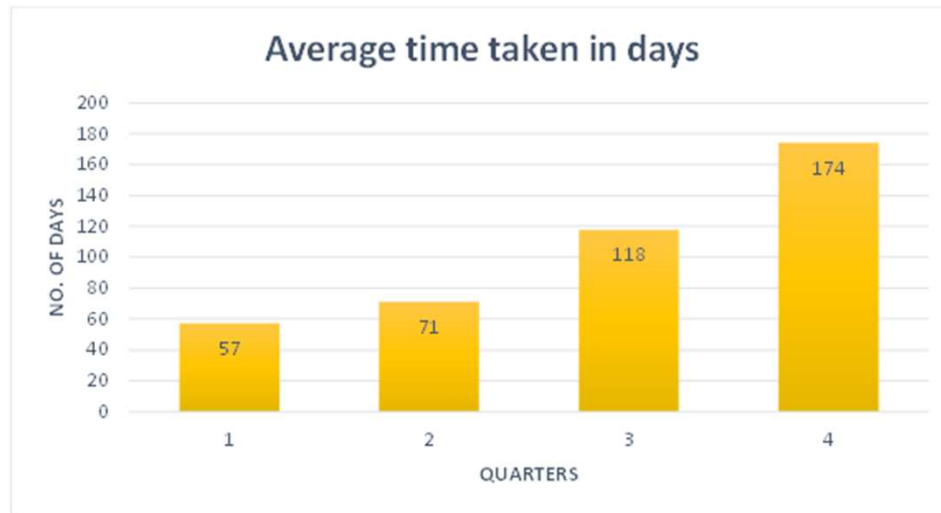


credit_card_type	avg_dis_rate
laser	64%
mastercard	63%
visa-electron	62%
china-unionpay	62%
americanexpress	62%
maestro	62%
instapayment	62%
jcb	61%
switch	61%
diners-club-carte-blanche	61%
bankcard	61%
diners-club-us-ca	61%
diners-club-enroute	60%
visa	60%
solo	59%
diners-club-international	58%

OBSERVATIONS/FINDINGS

- There are 16 different types of credit cards.
- Among them, Laser offers more discounts than any other type of card.
- Master and Visa-electron followed with 63% and 62% after Laser.
- Diners-club-international and Solo offered the least discount rates.
- Just a small percentage change in the average discount rate of each credit card type, ranging from 58% to 64%.

TIME TAKEN TO SHIP ORDERS BY QUARTER



quarter_number	avg_days
1	57
2	71
3	118
4	174

OBVERSATIONS/FINDINGS

- Average time taken to ship orders is 57 days in quarter 1, 71 days in quarter 2, 118 days in quarter 3, and 174 days in quarter 4, respectively.
- The average time take to ship orders is increasing from quarter to quarter.
- There is a 117-day difference in shipping time when comparing quarter 1 to quarter 4.

INSIGHTS & RECOMMENDATIONS

INSIGHTS

- There are a total of 994 customers across 45 states, with California, Texas, Florida, New York and District of Columbia having the most customers.
- The average customer rating in quarter 1 was 3.55; however, it declined to 2.40 in quarter 4. The average rating has been decreasing from quarter to quarter.
- In quarter 1, the overall positive feedback from customers was around 58.71%, with negative feedback at only 22.26%. However, in quarter 4, positive feedback decreased to 20.1% while negative feedback increased to 59.8%. This indicates a growing dissatisfaction among customers over time.
- Our data shows that that Chevrolet, Ford, Toyota, Pontiac and Dodge are the most preferred vehicle makers by customers overall. Specifically, in many states, Chevrolet, Dodge, Ford are consistently the most preferred vehicle makers.

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INSIGHTS & RECOMMENDATIONS

INSIGHTS

- There are a total of 1,000 orders across four quarters, with 310 orders placed in quarter 1 and 199 orders placed in quarter 4. There is a decreasing trend in purchasing from quarter to quarter.
- The total revenue across four quarters is 82.6 million. While the revenue was strong in quarter 1 at 26 million, it declined to 15 million in quarter 4. Furthermore, the revenue percentage changes, from -18.61% in quarter 2 to -22.47% in quarter 4, indicate a consistent decrease in revenue each quarter.
- There are 16 different types of credit card types, and average discount rates range from 58% to 64%. Among them, Laser offers the highest rate at 64%.
- The average shipping time from order placement is 57 days in quarter 1 and 174 days in quarter 4. Across all quarters, the average delivery time is 105 days.



INSIGHTS & RECOMMENDATIONS

RECOMMENDATIONS (1)

- Customer satisfaction is extremely important in every business, as it has ability to make or break your business. Thus, we need to identify the root causes of growing customer dissatisfaction.
- There are many possible factors that can make customers dissatisfied, such as poor sales service, issues with the financing process, and inadequate after-sale service. The best way to identify these issue is by using customer feedback surveys.
- After identifying the issues, we need to take corrective actions professionally and implement preventative measures to avoid repetitive problems.



INSIGHTS & RECOMMENDATIONS

RECOMMENDATIONS (2)

- According to the current consumer trend, individuals conduct pre-purchase research before making a buying decision. Poor ratings and negative reviews have the potential to turn customers away immediately, leading to a loss of revenue and undermining the business's reputation. To address this issue, we should employ the following techniques:
 - ✓ **Promptly Respond to Poor Ratings and Negative Reviews:** Ignoring negative feedback can be detrimental. Respond to poor ratings and negative reviews online as quickly as possible.
 - ✓ **Resolve Issues Offline:** Work to solve issues before customers take them online. Continue the conversation offline and invite the customer to contact you directly to resolve the situation.
 - ✓ **Encourage Positive Reviews:** Actively encourage satisfied customers to leave positive reviews. This proactive approach can help offset negative reviews and enhance the overall rating of your business.



INSIGHTS & RECOMMENDATIONS

RECOMMENDATIONS (3)

- The consequences of customer dissatisfaction and poor ratings often lead to a decline in the total number of orders. In addition to addressing customer dissatisfaction issues, the following techniques should be practiced to boost sales:
- **Choose the Right Marketing Campaign:** Selecting the appropriate marketing campaign can significantly impact sales
- **Reach Out to Existing Customers:** Retaining old customers is often more cost-effective than acquiring newer ones. Establishing and maintaining a positive relationship with existing customers can lead to repeat business.
- **Optimize Business Website for Search Engines:** Ensure that your business website is optimized for search engines to increase visibility and attract potential customers.
- **Analyze Pricing and Be Competitive:** Regularly analyze pricing strategy to stay competitive in the market and attract price-sensitive customers.



INSIGHTS & RECOMMENDATIONS

RECOMMENDATIONS (4)

- Timely delivery is crucial for business success, as customers expect their orders to be delivered quickly and efficiently, without any hassle. To enhance delivery time, it's necessary to first identify the bottlenecks in the supply chain and maintain effective communication with customers and suppliers. Recommendation include:
- Conducting a Through Analysis: A through analysis of the delivery process can help pinpoint weak links and areas that need improvement.
- Tracking Time for Each Step: Track the time it takes for each step in the delivery process. By analyzing this data, we can identify which steps are taking the longest and where delays are occurring.
- Maintaining Clear and Consistent Communication: Maintain clear and consistent communication with customers and suppliers at every state of the delivery process. Provide accurate tracking information, offer proactive updates on any delays, and handle customer queries promptly.

INSIGHTS & RECOMMENDATIONS

RECOMMENDATIONS (5)

- Lastly, adopt the principle of “giving customers what they want, not what you think they need.” With data indicating that states with the most customers and their preferred vehicle makers, it’s crucial to stock the right vehicles intelligently.
- This strategic approach not only reduce floorplan fees but also minimizes waiting time by delivering customers their most preferred vehicles from other states. This is not only enhancing profitability but also ensures customer satisfaction.





THANK YOU!